

Set Up Proxmox and Ansible; Terraform

1. Proxmox Installation on the Server

Prerequisites

- Server hardware: ensure it's compatible (64-bit CPU with Intel VT/AMD-V support, at least 2GB RAM, and 16GB storage).
- Proxmox ISO image: download the latest from Proxmox's official website.

Installation Steps

1. **Boot from ISO:** Insert USB or mount ISO, boot the server, and select the Proxmox installation.
2. **Installation Wizard:**
 - Accept the license agreement.
 - Select the target disk (ensure you choose the correct one, as it will format the disk).
 - Configure network settings, hostname, and admin credentials.
3. **Initial Setup:**
 - Once installed, access Proxmox from a browser at `https://<Server_IP>:8006`.
 - Log in with root and the password set during installation.

2. Initial Configuration of Proxmox

- **Network Configuration:** Set up a static IP and any necessary VLAN configurations.
- **Storage Configuration:** Define any additional storage (local disks, NFS, or iSCSI) for use with Proxmox.
- **Update Repositories & Install Vim:** bash Copy code `apt update && apt upgrade -y && apt install vim`

3. Preparing Proxmox for Ansible and Terraform Integration

- **Enable SSH Access:**
 - Set up SSH access for Ansible and Terraform to interact with Proxmox.
 - Generate SSH keys (if not done) and configure key-based access to streamline automation.
 - [Setup SSHD For Remote Access](#)
- **Install Proxmox API Access Tools:**

- The `pve-api` tool may be useful for direct API calls.
- **Install Necessary Proxmox Modules** for integration:
 - Install `python3-pvea` or similar packages on your Ansible/Terraform controller for managing the Proxmox environment.

4. Ansible and Terraform Configuration

Ansible Setup

- **Proxmox Inventory:** Create a basic Ansible inventory file that lists Proxmox nodes.
- **Playbooks:** Write Ansible playbooks to automate VM creation and template preparation for Kasm containers.
- **Proxmox Modules:** Use the `proxmox_kvm` module to create, configure, and manage VMs.

Terraform Setup

- **Terraform Provider for Proxmox:**
 - Install the `hashicorp/proxmox` provider plugin.
 - Set up `main.tf` to manage infrastructure, define VM templates, and network configuration.

5. Deploying Kasm Containers with Preconfigured OS Templates

- **Install Kasm on Proxmox:**
 - Use an Ansible playbook or manually deploy Kasm via Docker on an Ubuntu container.
 - Configure networking for Kasm workspaces.
- **Automate VM and Container Provisioning:**
 - Use Terraform to deploy various VM templates (Kali, Ubuntu, Windows).
 - Configure network bridging and firewalls to allow VM/container access.

6. Testing and Validation

- Run tests to confirm successful Kasm container operation and connectivity.
- Ensure Ansible playbooks and Terraform plans can recreate the environment without issues.